PHYSICS DEPARTMENT
Rabih A.G.

Chapter 16: Capacitors

Practical Application Worksheet

Instructions: Circle the correct answer for each multiple-choice question. Use the blank space provided below each question to show your calculation steps.

1. What are the two parallel conducting plates of a capacitor commonly called, and what type of material separates them?

A. Armatures (conductors) separated by a dielectric (insulator).
B. Dielectrics (insulators) separated by an armature (conductor).
C. Terminals (insulators) separated by a vacuum.
D. Armatures (insulators) separated by a dielectric (conductor).

2. A capacitor has a capacitance of $5\,\mu F$ and is connected to a $12\,V$ battery. What is the total charge stored on its positive plate?

A. $2.4\,\mu C$
B. $17\,\mu C$
C. $60\,\mu C$
D. $600\,\mu C$

3. How much electrostatic energy is stored in a $10\,\mu F$ capacitor that is fully charged to $20\,V$?

A. $100\,\mu J$
B. $200\,\mu J$
C. $2\,mJ$
D. $4\,mJ$

4. A parallel-plate air capacitor has square plates of side length 10 cm and a separation distance of 2 mm. Given the permittivity of free space $\epsilon_0 \approx 8.85 \times 10^{-12}$ F/m, what is its approximate capacitance C_0? (Note: $\epsilon_r = 1$ for air).

A. 8.85 pF
B. 22.1 pF
C. 44.25 pF
D. 4.42 nF

5. A parallel-plate air capacitor has a capacitance of C_0. If the air is replaced entirely by mica, which has a relative permittivity of $\epsilon_r = 5.4$, what is the new capacitance?

A. $C_0/5.4$

B. $C_0 - 5.4$

C. $C_0 + 5.4$

D. $5.4C_0$

6. Two capacitors, $C_1 = 4\,\mu F$ and $C_2 = 12\,\mu F$, are connected in series. What is the equivalent capacitance of this combination?

A. $3\,\mu F$

B. $8\,\mu F$

C. $16\,\mu F$

D. $48\,\mu F$

7. A circuit requires an equivalent capacitance of $15\,\mu F$. If you already have a $10\,\mu F$ capacitor installed, what component should you connect in parallel with it to reach the target?

A. A $2.5\,\mu F$ capacitor

B. A $5\,\mu F$ capacitor

C. A $30\,\mu F$ capacitor

D. A $150\,\mu F$ capacitor

8. An RC circuit is built using a $20\,k\Omega$ resistor and a $50\,\mu F$ capacitor. What is the time constant (τ) of this circuit?

A. $0.1\,s$

B. $1.0\,s$

C. $2.5\,s$

D. $1000\,s$

9. An completely uncharged capacitor is connected in series with a resistor and a $10\,V$ power supply. What is the approximate voltage across the capacitor at exactly $t = \tau$?

A. $3.7\,V$

B. $5.0\,V$

C. $6.3\,V$

D. $10.0\,V$

10. A capacitor is fully charged to $24\,V$. The power supply is disconnected, and the capacitor begins to discharge through a resistor. What will be the approximate voltage across the capacitor at $t = \tau$?

A. $0\,V$

B. $8.9\,V$

C. $12.0\,V$

D. $15.1\,V$

11. If the distance d between the two plates of a parallel-plate capacitor is halved while keeping the overlapping area constant, how does the capacitance change?

A. It halves.

B. It remains unchanged.

C. It doubles.

D. It quadruples.

12. A capacitor stores an electrostatic energy W when connected to a potential difference U. If the voltage applied to this exact same capacitor is doubled to $2U$, what is the newly stored energy?

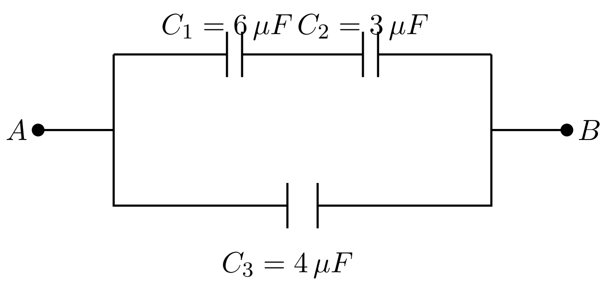
A. $0.5W$

B. W

C. $2W$

D. $4W$

13. Three capacitors are connected in a complex circuit as shown below. $C_1 = 6\,\mu F$ and $C_2 = 3\,\mu F$ are connected in series. This series branch is connected in parallel with a third capacitor, $C_3 = 4\,\mu F$. What is the total equivalent capacitance (C_{eq}) between points A and B?



A. $1.33\,\mu F$

B. $2.4\,\mu F$

C. $6.0\,\mu F$

D. $13.0\,\mu F$

14. Refer to the circuit diagram from the previous question. If a constant DC voltage of $U_{AB} = 12\,V$ is applied across the entire combination, determine the charge Q_1, the voltage U_1, and the stored electrostatic energy W_1 of capacitor C_1.

A. $Q_1 = 24\,\mu C, \quad U_1 = 4\,V, \quad W_1 = 48\,\mu J$

B. $Q_1 = 72\,\mu C, \quad U_1 = 12\,V, \quad W_1 = 432\,\mu J$

C. $Q_1 = 24\,\mu C, \quad U_1 = 8\,V, \quad W_1 = 96\,\mu J$

D. $Q_1 = 48\,\mu C, \quad U_1 = 4\,V, \quad W_1 = 24\,\mu J$